

# A machine-verified erratum and its repair: Brown–Zudilin’s $\zeta(5)$ denominators and the Frobenius certificate, formalized in Lean 4

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## Abstract

Brown and Zudilin’s paper on cellular rational approximations to  $\zeta(5)$  [1] states, as an experimental observation, the integrality  $d_n^5 P_n \in \mathbb{Z}$  for the rational coefficients of its totally symmetric linear forms, where  $d_n = \text{lcm}(1, \dots, n)$ . The paper’s own displayed value  $P_2 = 1190161/384$  contradicts this inclusion, failing by a factor of exactly 12; the claim is still present, unamended, in the January 2026 revision. We present a Lean 4 formalization, mechanically verified by the Lean kernel with zero `sorry`s and only the three standard axioms, of: (i) the contradiction, with  $P_2$  *derived* from the construction rather than transcribed; (ii) its sharp repair at  $n = 2$ :  $c d_2^5 P_2 \in \mathbb{Z}$  if and only if  $12 \mid c$  (companion form: if and only if  $2 \mid c$ ); and (iii) the theorem that proves the corrected law on the clean interval — the *Frobenius certificate* theorem of [3]: for every  $n \geq 1$  and every prime  $p$  with  $5 \leq p$ ,  $n < p \leq 2n$ , the harmonic functionals satisfy  $w_n \equiv \tilde{w}_n \equiv 0 \pmod{p}$ , and hence  $p \mid p_n$ . The formalization exposed two statement-level errors in our own paper-level drafts, which we document. A second wave (Section 6) adds: the corrected law kernel-verified for  $n \leq 8$  with nonvanishing; a Lucas–Frobenius congruence  $Q_{ap+r} \equiv Q_a Q_r \pmod{p}$  for the cellular coefficient (whose formalization caught a gap in the original proof sketch); and the *conditional assembly*: the descent hypothesis (D) implies the full law at every prime  $p \geq 5$  — so the complete corrected law is now, formally, one explicit hypothesis away. The complete development builds in one `lake build` and is publicly available.

## 1 Introduction

The irrationality theory of odd zeta values rests on rational linear forms in zeta values whose coefficients must be cleared of denominators at a controlled exponential cost: for a weight-5 form the benchmark cost is  $d_n^5$ , with  $d_n = \text{lcm}(1, 2, \dots, n) \sim e^n$ . In the totally symmetric cellular construction of Brown and Zudilin [1], the linear form

$$I_n = Q_n(2\zeta(5) + 4\zeta(3)\zeta(2)) - 4\hat{P}_n \zeta(2) - 2P_n \quad (1)$$

carries rational coefficient ladders  $Q_n$ ,  $\hat{P}_n$ ,  $P_n$ , and the paper states (in the display labeled `dn-totsym`, presented explicitly as an experimental observation from “an extensive computation of the rational coefficients”):

$$Q_n, d_n^2 d_{2n} \hat{P}_n, d_n^5 P_n \in \mathbb{Z} \quad (n = 0, 1, 2, \dots). \quad (2)$$

The same paper displays (in the block following the definition of  $I_n$ ) the exact values

$$Q_2 = 2989, \quad \hat{P}_2 = \frac{344923}{96}, \quad P_2 = \frac{1190161}{384}. \quad (3)$$

Since  $d_2 = 2$  and  $384 = 2^7 \cdot 3$ , we have  $d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$ : the third inclusion of (2) fails already at  $n = 2$ , by a factor of exactly 12 (and the second fails by a factor of exactly 2:  $d_2^2 d_4 \hat{P}_2 = 344923/2$ ). The corrected law

$$12 d_n^5 P_n \in \mathbb{Z}, \quad (4)$$

conjecturally with the  $\{2, 3\}$ -ceilings sharp, has substantial computational support, and its clean-interval part is a theorem: the *Frobenius certificate* theorem of the companion paper [3] shows that every prime  $n < p \leq 2n$  (the primes of  $\binom{2n}{n}$ , which the naïve partial-fraction bound admits into the denominator) in fact divides the numerator  $p_n$ .

This note reports a complete formalization of the erratum, its sharp  $n = 2$  repair, and the clean-interval theorem in Lean 4 over Mathlib. Every theorem below is verified by the Lean kernel; the development contains zero `sorry`s, uses no `native_decide`, adds no axioms, and each named theorem reports exactly the standard axiom triple `[propext, Classical.choice, Quot.sound]`. The trust base is the Lean kernel and the printed displays of [1] quoted above — nothing else.

Two purposes motivate machine verification here. First, an erratum to a published paper should meet a higher evidentiary standard than the claim it corrects; a kernel-checked derivation of  $P_2$  from the construction, ending in a formal negation of (2) at  $n = 2$ , is the strongest form of such a statement we know how to produce. Second, the formalization is a test of a design thesis: proofs shaped as *finite certificates plus one small uniform identity* formalize at low cost. The clean-interval theorem has exactly this shape, and the entire development — definitions, numeric gates, five lemma modules, assembly — was completed in a single day. As a byproduct the formalization caught two statement-level errors in our own drafts (Section 5), neither of which affects the mathematics, both of which would likely have survived human refereeing.

## 2 The construction and the formal objects

Fix  $n \geq 1$  and set

$$R_n(k) = (n!)^4 \left(k + \frac{n}{2}\right) \frac{\prod_{j=1}^n (k-j) \prod_{j=1}^n (k+n+j)}{\prod_{j=0}^n (k+j)^6}, \quad (5)$$

Zudilin’s totally symmetric rational function [2]. Its partial fractions

$$R_n(k) = \sum_{i=1}^6 \sum_{j=0}^n \frac{a_{i,j}}{(k+j)^i}, \quad \tilde{a}_{i,j} = j(n-j) a_{i,j} + (2j-n) a_{i+1,j} - a_{i+2,j}$$

(indices above 6 read as zero; the  $\tilde{a}$  are the partial fractions of  $-k(k+n)R_n(k)$ ) define the functionals

$$u = \sum_j a_{5,j}, \quad w = \sum_j a_{3,j}, \quad v = \sum_{i,j} a_{i,j} H_j^{(i)}, \quad H_j^{(i)} = \sum_{m=1}^j m^{-i},$$

their tilded companions, and the determinant ladders

$$q_n = u\tilde{w} - \tilde{u}w, \quad p_n = \tilde{w}v - w\tilde{v}, \quad \tilde{p}_n = u\tilde{v} - \tilde{u}v. \quad (6)$$

The Brown–Zudilin coefficients of (1) correspond via  $P_n = (-1)^{n+1} p_n / \binom{2n}{n}$  (the display following (2) in [1], which ties the cellular normalization to [2]).

In Lean, all of these are *computable* functions on  $\mathbb{Q}$ : the primary definition of  $a_{i,j}$  is the truncated local Taylor expansion of  $B_j(k) = (k+j)^6 R_n(k)$  at  $k = -j$  (module `Cbcert/Defs.lean`), and the partial-fraction property is then a *theorem* (`pf_cleared`, module `Cbcert/PartialFraction.lean`): the cleared polynomial identity

$$N_n(k) = \sum_{i,j} a_{i,j} (k+j)^{6-i} \prod_{m \neq j} (k+m)^6,$$

proven for all  $n$ . Computability is what makes the development self-auditing: a numeric gate (`Cbcert/Numeric.lean`) reduces the full definition chain by kernel computation at  $n = 2$ ,

$$w = \frac{6125}{4}, \quad \tilde{w} = 1764, \quad u = 469, \quad \tilde{u} = 552, \quad v = \frac{74463}{32}, \quad \tilde{v} = \frac{43085}{16}, \quad p_2 = \tilde{w}v - w\tilde{v} = -\frac{1190161}{64},$$

and at  $n = 3$ , in exact agreement with the independent reference implementation; the frozen definitions were gated on this agreement before any lemma work began. Note  $P_2 = (-1)^3 p_2 / \binom{4}{2} = 1190161/384$ , matching (3) exactly.

### 3 The machine-verified erratum

**Theorem 1** (`Cbcert/ErrorExhibit.lean`). *Define  $P_2$  by the construction of Section 2 (derived, not transcribed). Then, verified by the Lean kernel with standard axioms:*

- (i) *`P2_eq`:  $P_2 = 1190161/384$ , and likewise  $Q_2 = 2989$ ,  $\hat{P}_2 = 344923/96$  — the construction reproduces all three displayed values (3) of [1].*
- (ii) *`bz_inclusion_fails`:  $d_2^5 P_2 \notin \mathbb{Z}$ . In particular the inclusion  $d_n^5 P_n \in \mathbb{Z}$  of (2) is false at  $n = 2$ .*
- (iii) *`clearing_iff`: for every  $c \in \mathbb{N}$ ,  $c d_2^5 P_2 \in \mathbb{Z} \iff 12 \mid c$ . The repair factor 12 of (4) is sharp at  $n = 2$ .*
- (iv) *`companion_inclusion_fails`, `companion_clearing_iff`:  $d_2^2 d_4 \hat{P}_2 \notin \mathbb{Z}$ , and  $c d_2^2 d_4 \hat{P}_2 \in \mathbb{Z} \iff 2 \mid c$ .*

The Lean statements are short enough to read in full (unicode transliterated):

```
theorem bz_inclusion_fails :
  ¬ ∃ m : ℤ, (d 2 : ℚ)^5 * P2 = (m : ℚ)
theorem clearing_iff (c : ℕ) :
  (∃ m : ℤ, (c : ℚ) * (d 2)^5 * P2 = m) ↔ 12 ∣ c
```

where `d` is the genuine running  $\text{lcm}(1, \dots, n)$ . Each definition and theorem carries a docstring citing the source labels of [1] (`dn-totsym` for the claim, the `In` display block for the values), so a reader can hold the PDF next to the Lean file. Two remarks on scope. The claim (2) is presented in [1] as an experimental observation, not a theorem; what is exhibited here is a false printed claim and its sharp correction, not a broken proof. And the bridge between the cellular normalization of [1] and the construction of Section 2 is the paper's own stated correspondence to [2]; it is corroborated *inside* the formalization by Theorem 1(i), which reproduces all three displayed values simultaneously from one derivation chain.

## 4 The Frobenius-certificate theorem, formalized

The mathematical content, proven in [3]: for a prime  $p$  with  $n < p \leq 2n$ , the Frobenius identity  $k^p - k \equiv (k + j)^p - (k + j) \pmod{p}$  makes  $\varphi = (k^p - k)^2$  congruent to  $(k + j)^2$  modulo  $(k + j)^6$  at every pole  $-j$ , while  $\deg \varphi = 2p \leq 4n + 2$  keeps  $\varphi R_n$  decaying at infinity; the residue-sum theorem on  $\mathbb{P}^1$  then forces  $\sum_j a_{3,j} \equiv 0 \pmod{p}$ . In certificate form this is the three-term relation with coefficients  $(c_3, c_{p+2}, c_{2p+1}) = (1, -2, 1)$  among the decay relations

$$E_M = \sum_{i=1}^{\min(6,M)} \binom{-i}{M-i} \sum_j j^{M-i} a_{i,j} = 0 \quad (1 \leq M \leq 4n+4),$$

which are the coefficients of the expansion of  $R_n$  at infinity.

Conceptually: the certificate must prescribe a 6-jet at each of the  $n + 1$  poles ( $6(n + 1)$  linear conditions) under a degree budget of  $4n + 3$  imposed by the residue theorem — an overdetermined Hermite interpolation problem in characteristic zero, which is why the characteristic-zero  $E_M$  relations alone cannot force the congruence. Frobenius solves it in characteristic  $p$  at degree  $2p$ , within budget precisely on the prime window; the mechanism is *Frobenius-assisted global jet interpolation plus residue duality*, and it generalizes to arbitrary polynomial weights  $Q$  with  $2p + \deg Q \leq 4n + 3$  [3].

**Theorem 2** (Cbcert/Main.lean). *For every  $n \geq 1$  and every prime  $p$  with  $5 \leq p$  and  $n < p \leq 2n$ , verified by the Lean kernel with standard axioms:*

$$\text{res\_congruence\_w} : p \mid w_n, \quad \text{res\_congruence\_wt} : p \mid \tilde{w}_n, \quad \text{res\_congruence\_pn} : p \mid p_n,$$

where  $p \mid x$  for a  $p$ -integral rational  $x$  means  $\text{res}_p(x) = 0$  under the residue homomorphism  $\text{res}_p$  to  $\mathbb{Z}/p$ . Equivalently (*pn\_valuation* etc.):  $p_n = 0$  or  $\text{ord}_p(p_n) \geq 1$ .

The last statement of Theorem 2 is the clean-interval central-binomial cancellation (CB<sub>1</sub>): every prime of the window  $(n, 2n]$ ,  $p \geq 5$ , divides  $p_n$ .

### Architecture

The proof formalizes in four independent modules plus an assembly, mirroring [3]:

| module               | content                                          | status        |
|----------------------|--------------------------------------------------|---------------|
| Defs.lean            | all definitions (computable over $\mathbb{Q}$ )  | frozen, gated |
| Certificate.lean     | Lemma B: the $\mathbb{F}_p$ certificate identity | proven        |
| PartialFraction.lean | pf_cleared: the all- $n$ cleared identity        | proven        |
| Decay.lean           | Lemma A: $E_M(a) = E_M(\tilde{a}) = 0$           | proven        |
| Integrality.lean     | Lemma C: $p$ -integrality of $a, \tilde{a}, H$   | proven        |
| Main.lean            | residue map, reordering, main theorems           | proven        |
| Numeric.lean         | kernel-reduction gates at $n = 2, 3$             | proven        |
| ErrorExhibit.lean    | Theorem 1                                        | proven        |

Lemma B is pure binomial arithmetic in  $\mathbb{Z}/p$ : for  $p \geq 5$ ,  $1 \leq i \leq 6$  and every  $x \in \mathbb{Z}/p$ ,

$$\sum_{M \in \{3, p+2, 2p+1\}} c_M \binom{-i}{M-i} x^{M-i} = \delta_{i,3} \quad \text{in } \mathbb{Z}/p,$$

proven from  $x^p = x$  and  $p \mid \binom{p}{m}$  for  $0 < m < p$ . It is the mathematically novel part of [3] and was the *easiest* module to formalize — a data point for the certificate-shaped design thesis.

*Lemma A* never needed the anticipated Laurent-series machinery: with the cleared identity `pf_cleared` in hand, the decay relations become the statement that  $E_M(a)$  is the coefficient of  $X^M$  in  $R_n(1/X) = X^{4n+5} \tilde{N}(X)/\tilde{D}(X)$  with  $\tilde{D}(0) = 1$ , an elementary computation in  $\mathbb{Q}[[X]]$ . The load-bearing insight in `pf_cleared` itself was that the truncated-multiplication primitive used by the computable definitions *is* the Cauchy product, so the Taylor-coefficient definition of  $a_{i,j}$  composes directly into the polynomial identity.

*Lemma C* bounds every denominator of  $a_{i,j}$ ,  $\tilde{a}_{i,j}$  ( $j \leq n$ ) and  $H_j^{(i)}$  by primes  $\leq n < p$ : the pole differences  $|j - j'| \leq n$  are the only sources.

*Assembly.* A residue homomorphism  $\text{res}_p$  is developed on the  $p$ -integral subring of  $\mathbb{Q}$  (compatible with  $+$ ,  $\times$ , and with  $\text{res}_p(x) = 0 \iff x = 0 \vee \text{ord}_p(x) \geq 1$ ), and a coefficient-agnostic reordering lemma turns Lemmas A + B + C into  $\text{res}_p(\sum_j b_{3,j}) = 0$  for both coefficient systems  $b \in \{a, \tilde{a}\}$  at once;  $p \mid p_n$  then follows from (6) by ring arithmetic. All three certificate indices satisfy  $M \leq 4n + 4$  (respectively  $4n + 2$  for  $\tilde{a}$ ) precisely because  $p \leq 2n$ ; the case  $p = 2n + 1$  is genuinely excluded.

## 5 What the formalization caught

Two statement-level errors in our own paper-level drafts were discovered and repaired during formalization. We record them because both are invisible to ordinary numerical testing on the intended domain, and both concern the *statements*, not the proofs.

- (1) **The valuation phrasing is unsound at zero.** The intended statement  $\text{ord}_p(p_n) \geq 1$  was originally frozen as  $1 \leq \text{padicValRat } p (p_n)$ . Mathlib defines `padicValRat`  $p \ 0 = 0$ , so this phrasing silently asserts  $p_n \neq 0$  — a genuinely separate arithmetic fact (true numerically for  $n \leq 60$ ; open in general, and *not* needed for any downstream denominator application). The canonical theorems are therefore stated in residue form, with the faithful disjunction  $p_n = 0 \vee 1 \leq \text{padicValRat } p (p_n)$  provided as a corollary. General nonvanishing is documented as an explicitly open problem, not a hidden hypothesis.
- (2) **The integrality lemma needs its hypotheses.** As first staged, Lemma C quantified over all pole indices  $j$  and all primes  $p > n$ . Both restrictions matter: the formal coefficient function evaluated out of range or at  $p = 2$  produces genuine counterexamples (at  $n = 1$ :  $a_{5,0} = 9/2$ , and the out-of-range  $a_{5,2} = 9/128$  with  $\text{ord}_2 = -7$ ). The proven lemma carries  $j \leq n$  and  $p$  odd; the false general wrappers were deleted.

Neither finding affects [3], whose arguments always operate in range with  $p \geq 5$ ; both are exactly the kind of boundary defect that survives refereeing.

## 6 The second wave: finite range, the Lucas–Frobenius row, and the conditional assembly

Three further results were formalized on 19–20 July 2026, extending the development from “the band theorem plus the erratum” toward the full corrected law.

## The corrected law for $n \leq 8$ , with nonvanishing

Modules `Cbcert/FiniteLaw/N2..N8.lean` verify, by pure kernel computation (no `native_decide`), the full corrected law  $12 d_n^5 P_n \in \mathbb{Z}$  for every  $n \leq 8$ , together with  $p_n \neq 0$  at each such  $n$  — so on this range the disjunctive main theorems collapse to the sharp valuation form unconditionally. An exact-arithmetic gate confirmed the law out to  $n = 24$  in Python before any Lean statement was generated, and each Lean file checks a precomputed integer witness rather than searching. (The kernel cost grows  $\approx 1.75\times$  per level and does not parallelize within a level;  $n \leq 8$  was chosen deliberately, and the independent adversarial sweep described below now covers  $n \leq 360$ .)

## The Lucas–Frobenius row

For the manifestly integral Brown–Zudilin double-binomial coefficient

$$Q_n = \sum_{k,l=0}^n \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n},$$

module `Cbcert/LucasRow.lean` proves, sorry-free with standard axioms:

```
theorem Q_lucas_frobenius (a r p : ℕ) (hp : p.Prime) (hr : r < p) :
  (Q (a*p + r) : ℤ) ≡ (Q a : ℤ) * (Q r : ℤ) [ZMOD p]
```

This is the “ $q$ -row” of the conjectural Frobenius structure on the period lattice: the diagonal of the descent, proven unconditionally. The proof uses Mathlib’s Lucas theorem (`Choose.choose_modEq_choose_mod_mul`) per binomial factor. Formalization again caught a genuine gap in the human proof sketch: on summands with a base- $p$  carry, the five per-factor Lucas splits do *not* hold simultaneously, so one cannot simply multiply five congruences; the correct argument exhibits a single common vanishing factor on the carry branch. The sketch was right in spirit and wrong in mechanism — precisely the kind of defect that survives refereeing.

## The conditional assembly: one hypothesis from the full law

Module `Cbcert/Assembly.lean` formalizes the master reduction: define

```
def DescentHyp : Prop :=
  ∀ (m p : ℕ), p.Prime → 5 ≤ p → p ≤ m →
    padicValRat p (P (m / p)) - 5 ≤ padicValRat p (P m)
```

— exactly the descent (D): one base- $p$  digit down costs at most five  $p$ -adic orders. Then, sorry-free with standard axioms:

```
theorem descent_law (hD : DescentHyp) :
  ∀ n ≥ 1, ∀ p prime, 5 ≤ p → (−5) · (Nat.log p n) ≤ padicValRat p (P n)
```

The proof iterates (D) down the digit chain and discharges both terminal cases from theorems already in the development:  $2n_L < p$  via the integrality lemmas (Lemma C), and  $n_L < p \leq 2n_L$  via the certificate theorem (Theorem 2) plus a one-carry Kummer lemma. Two design points matter. First, the statement is *trap-free*: under Mathlib’s `padicValRat p 0 = 0` convention the induction closes with no nonvanishing hypotheses anywhere. Second, `hD` is a *hypothesis*, not an axiom — the axiom audit shows the standard triple only, and the theorem states formally and exactly what

remains open: prove (D), and the full corrected law at every  $p \geq 5$  follows by composition of kernel-verified pieces.

(D) itself is the object of an active proof campaign (with the collaborating model Sol, a GPT-5.6 instance, whose sustained multi-session work produced the entire Phase 2 reduction summarized here and in the companion progress note), which has reduced it, on paper, through a chain of exact certificate-shaped reductions — per-digit gates at the true singularities of the underlying Apéry-like recurrence, then a proved  $a = 1$  midpoint gate, with the remaining difficulty localized to a single period connection coefficient — each step verified in exact arithmetic before use. The adversarial evidence base for (D) now stands at  $n \leq 360, p \leq 73$ : 5,989 exact descents, zero violations, including the previously untested three-digit descent ranges, with the law riding the boundary (slack 0) at  $p = 2$  for every  $n$  — the constant 12 is exactly sharp throughout. Finite evidence, not proof; but the formal statement of what it is evidence *for* is now frozen in **DescentHyp**.

## 7 Scope and open problems

The formalized results are: the erratum and its sharp  $n = 2$  repair (Theorem 1); the clean-interval theorem (Theorem 2); the finite-range law with nonvanishing for  $n \leq 8$ ; the Lucas–Frobenius row; and the conditional assembly (D)  $\Rightarrow$  full law at  $p \geq 5$  (Section 6). Explicitly *not* formalized, because they are not yet theorems on paper: the descent (D) itself (the single remaining gap, object of the active campaign described above); the  $\{2, 3\}$ -part of the constant 12 (the fixed losses at  $p = 2, 3$ , a separate finite obligation); and the general nonvanishing  $w_n, \tilde{w}_n, p_n \neq 0$  (not needed by anything above).

## 8 Verification

The development lives in the `cbcert/` directory of the public repository [6], pinned to Lean toolchain `v4.33.0-rc1` and a fixed Mathlib commit. To verify: `lake exe cache get && lake build` (the full build reports  $\approx 8680$  jobs and finishes with no `sorry` warnings), then `#print axioms` on any theorem named above — including `Q_lucas_frobenius`, `descent_law`, and the `FiniteLaw` aggregate `law_upto`; each reports exactly `[propext, Classical.choice, Quot.sound]` (in particular no `sorryAx` in `descent_law`: the descent enters as a named hypothesis, not an axiom). The repository README contains the recipe, including which displays of [1] to compare against which Lean docstrings.

### Provenance

The formalization was produced on 19 July 2026 in a single orchestrated session by a fleet of LLM agents (Claude, Anthropic), following the statement-first protocol described above: definitions frozen against exact numeric gates before lemma work, statements corrected only under the documented-erratum discipline of Section 5. As with any formal development, correctness rests on the Lean kernel and the pinned toolchain, not on the process that produced the proof scripts.

## References

- [1] F. BROWN, W. ZUDILIN, On cellular rational approximations to  $\zeta(5)$ , arXiv:2210.03391 [math.NT] (2022; revised 26 January 2026).

- [2] W. ZUDILIN, A third-order Apéry-like recursion for  $\zeta(5)$ , *Mat. Zametki* **72** (2002), no. 5, 796–800; English transl., *Math. Notes* **72** (2002), no. 5–6, 733–737.
- [3] A Frobenius certificate for central-binomial cancellation in Zudilin’s symmetric  $\zeta(5)$  construction, companion paper, this repository (`worthiness/cb_certificate.tex`), 17 July 2026; revised 19 July 2026.
- [4] L. DE MOURA, S. ULLRICH, The Lean 4 theorem prover and programming language, *CADE 28*, Lecture Notes in Comput. Sci. **12699** (2021), 625–635.
- [5] THE MATHLIB COMMUNITY, The Lean mathematical library, *CPP 2020*, ACM (2020), 367–381.
- [6] Odd zeta values: machine verifications (repository), <https://github.com/rain-1/odd-zeta-values-autoformalization>.